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PROJECT REPORT

Department of Electrical Engineering

Presented to
Dr. V. Logeeshan

Presented by

Ranasingha R.A.T.U.	- 230519F
Ranaweera R.K.D.H.M	- 230529K
Ranmuthu S.G	- 230532M
Rashmika R.R.P.	- 230538L
Rifnaz K.R.M.	- 230550P

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INTRODUCTION

Introduction to the project

The Advanced Light Intensity Indicator (ALII) module is a specially designed system that monitors ambient light levels and displays both the stable/instantaneous and average light intensity over time. The project relies on analog filtering, logic gates and flip flop based digital circuits without using programmable ICs in order to build a fully functional and simple light monitoring device.

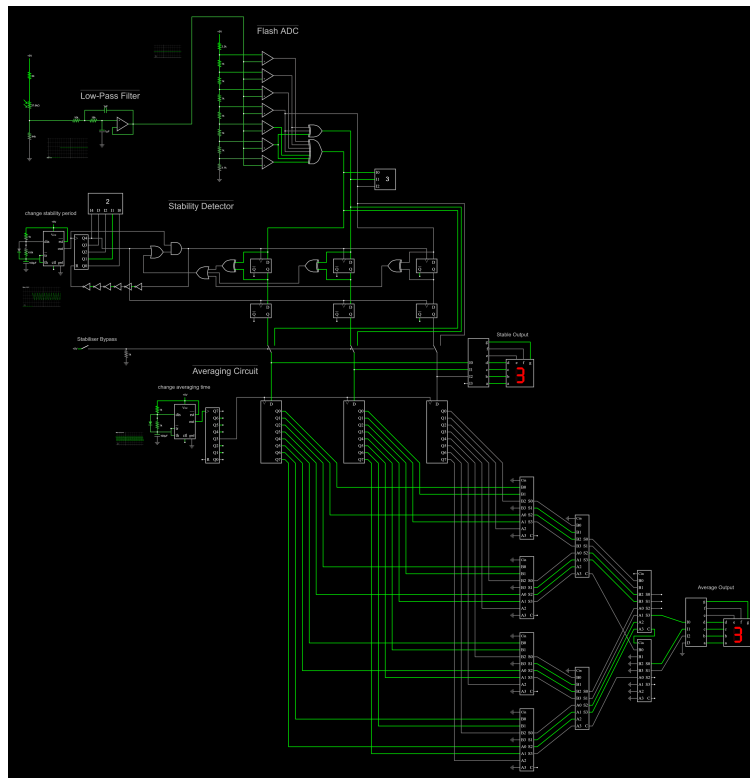
Purpose of the device

The primary purpose of the ALII module is to measure, process and indicate light intensity in a reliable and energy efficient way. The system allows users to monitor long term lighting conditions using an average intensity display. By converting LDR based light measurements into digital output levels the system ensures adequate lightning in public places and improved energy saving strategies

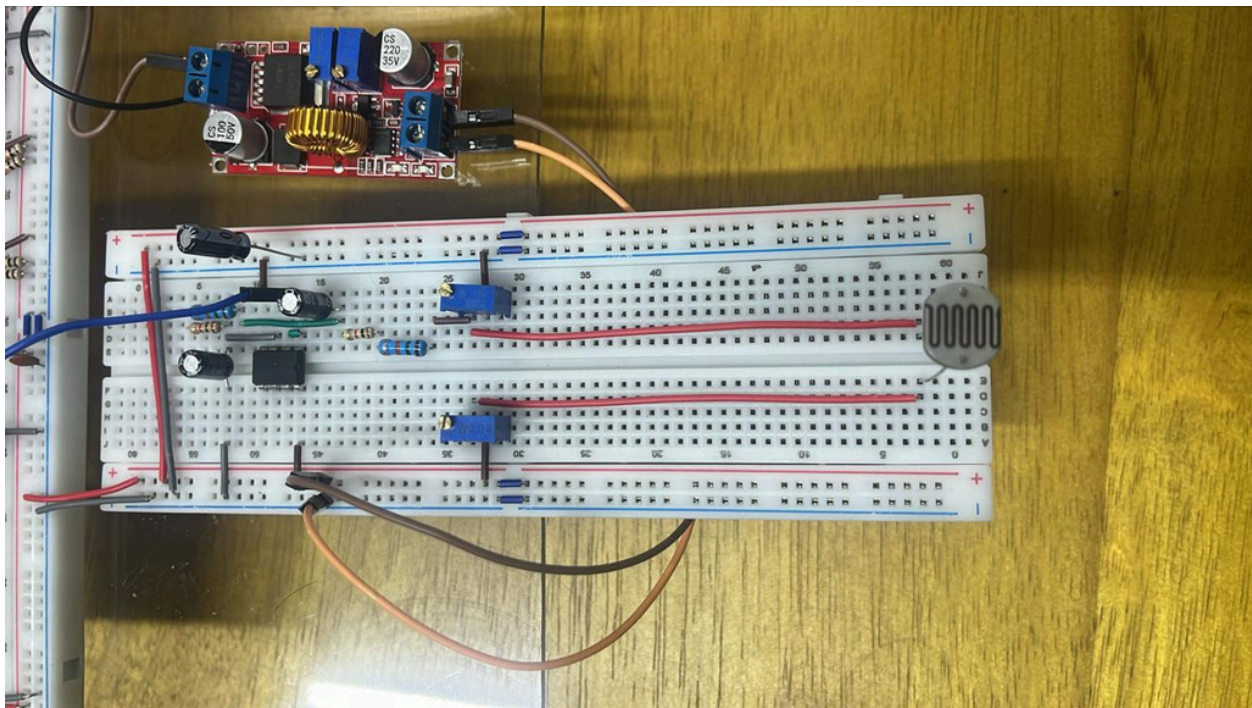
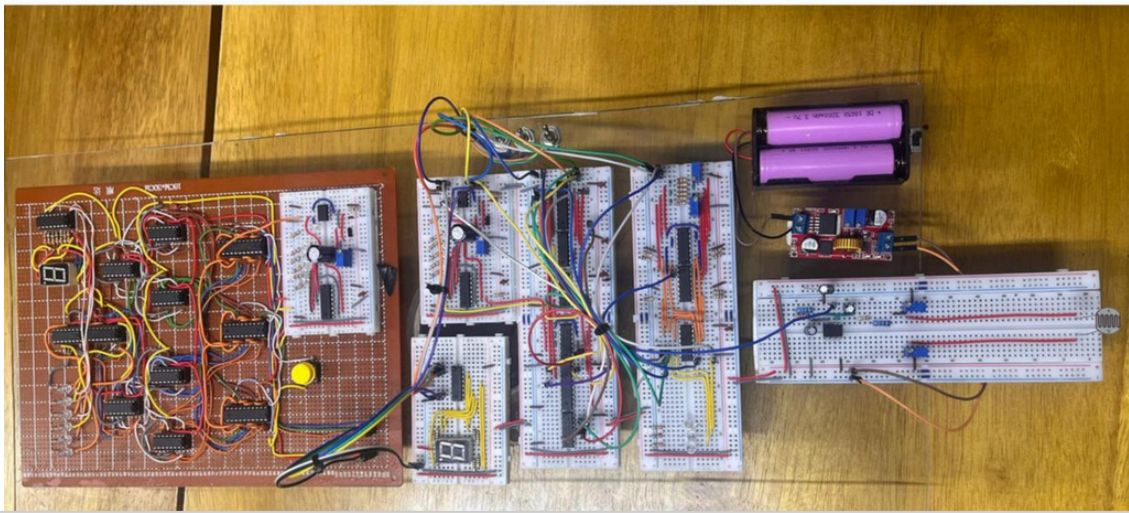
Parts of the device

1. LDR and filtering
 2. Flash ADC
 3. Stability detector
 4. Averaging part
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Simulation of the circuit :-

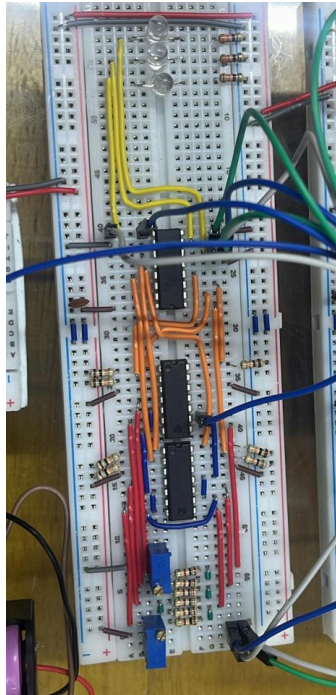


Physical circuit :-



To capture the changes of light intensity we use a LDR. During the light it decreases its resistance and during the darkness it increases the resistance. And it acts as a voltage divider. This voltage divider circuit is connected to the filter. We use a filter to eliminate unwanted electrical noise that may appear on the power lines supplying the LDR circuit (normally 50 - 100 Hz). For our purpose we chose a second order - sallen key low pass filter(using the MCP6002 op-amp). To set the cut off frequency as 20 - 50 Hz we selected $1\mu\text{F}$ capacitors and $16\text{k}\Omega$ resistors.

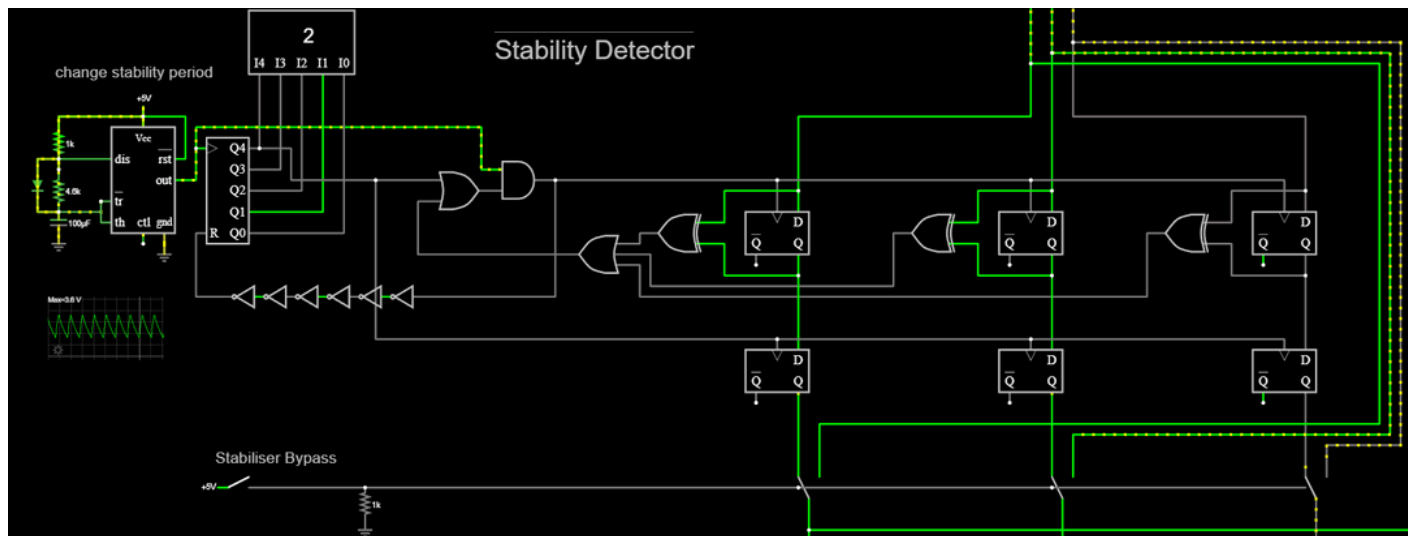
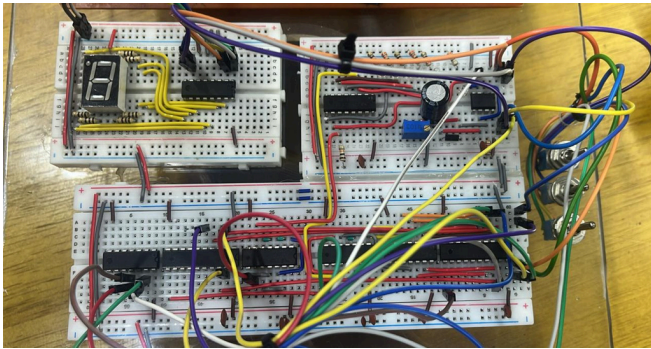
2. Flash ADC



A flash ADC is connected to the filter output and its purpose is to convert the analog signal into a digital signal. Since we were asked to display the intensity between 0-7 values, we use a 3 bit ADC type. For that we need 8 resistors and 7 comparators (LM339). 5V reference voltage is connected to the chain of resistors. For the first and last resistors we selected multi turn potentiometer to change the resistance to keep the range of seven segment from 0 to 7. Then the output of the comparators is connected to a 8 to 3 priority encoder(74LS148). It encodes the comparator output to 3bit binary number.

3.Stability detector

when stabilizer bypass switch is off,



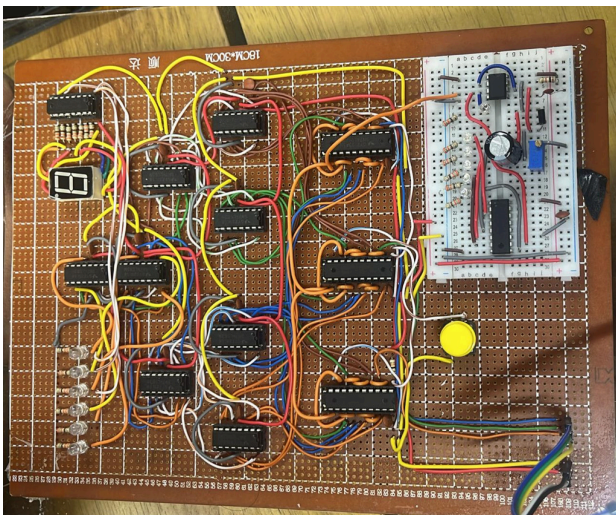
The stability detector is a circuit that outputs the input binary number only if it stays stable for a set period of time. If the input changes within that time period a timer is reset and it again waits for the set period while checking if the input changes. This is useful to filter out sudden variations in light intensity like a cloud passing through the sky.

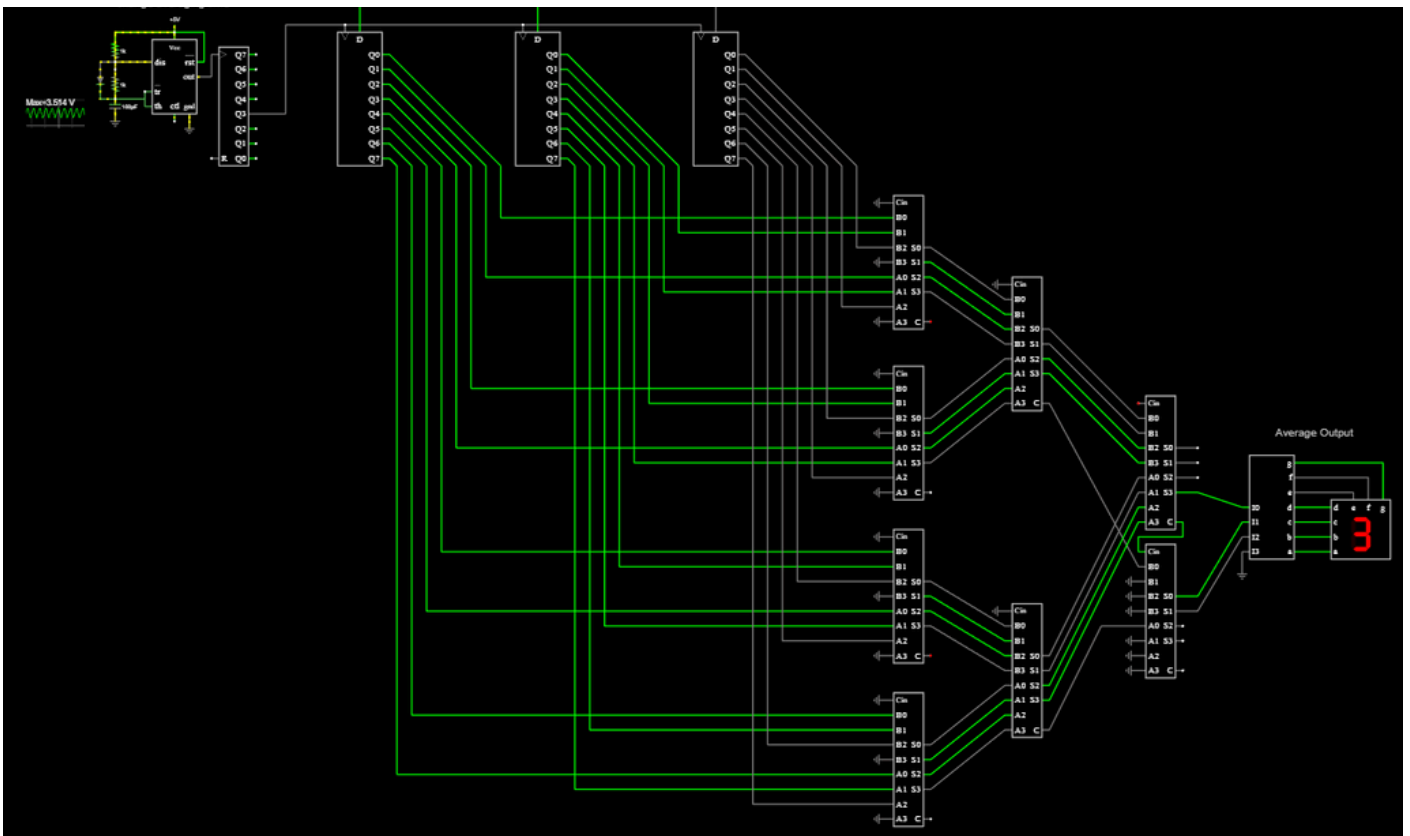
To set the time period of the stabilizer a 555 timer in Astable configuration and a binary counter is used. When the counter reaches a certain number (Eg: 16 or 32) a pulse is sent to the reset pin of the counter to restart the counter. Also, the same signal is used to send a clock pulse to 3 flipflops to output the stable output to the output pins.

The input comparison part is done with another set of 3 flipflops. When the clock resets a pulse is sent to these flipflops to store the current value of the input temporarily. Then this stored value is compared with the input value using three XOR gates. If one of the three bits change during the time period the output of the exor gates goes high and it is used as a signal to reset the clock, and as a clock pulse for the input flipflops to input the value at the input pins to the flipflops to compare again.

Also, a stabilizer bypass switch is added bypass the stabilizer if needed. Using bypass can take the actual value in filtererd signal.

3.Averaging part





The averaging part uses shift registers, adders, counters and seven segment decoder and display. When the stability detector confirms that the input signal is steady, the averaging process begins. Each new sample of the signal first enters the first 8 bit shift register(SN74LS273), and every time a new sample arrives, the older samples are pushed forward into the next shift registers. This creates the chain where the output of one shift register becomes the input of the next. All of the values stored inside these shift registers are then sent to an 4 bit binary full

Methods Used and Explanation

- Integrator Using

First we tried with Integrator in average part and stability part as analog circuit. But since leakage current from the circuit we decided to do that in digital method using Flip-Flops.

- Averaging methods

In averaging our method used to take a moving average , in first time we tried to take average with integrator but not a moving average. But for more accuracy we decided to move to moving average. And we found methods with accumulator and shift registers, and we thought using shift registers we can obtain our result properly.

Physical Construction

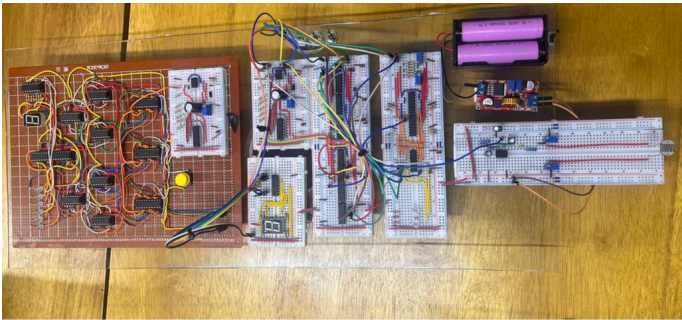
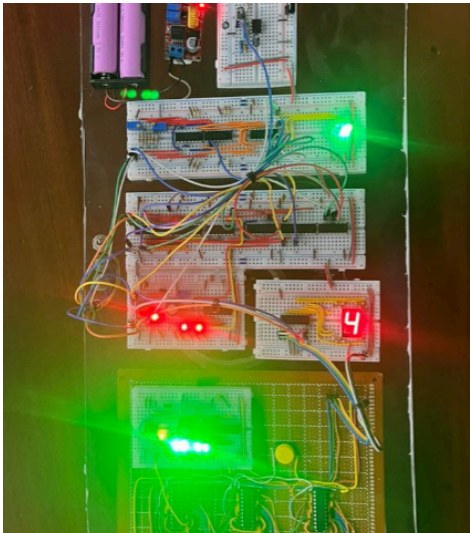
In physical construction of the circuit we planned to convert the breadboard circuit in to a PCB, but due to many reasons we had to make the full circuit using breadboard and dot boards. In dot board we could make a big circuit in a proper manner clearly.

References

<https://www.youtube.com/watch?v=utBQqcuOt9U>

https://youtu.be/iOSNb_dkYI?si=wkLOXvIpyOp6GJb2

<https://www.ti.com> - Texas Instruments



Thank You.
